

Loko

Binaural Space

User Manual - v1.0.0

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Thanks for using **Loko**.

Loko puts a sound somewhere around the listener's head - at a bearing, a height and a distance that can be pointed at, rather than somewhere on a line between two speakers.

It's played from one placement disc and five dials: **Azimuth**, **Elevation**, **Distance**, **Width** and **Gain**. A **3D** switch takes the vertical axis out, and a **LIMITER** lamp glows while the output is being held down. It works best on headphones.



Why a model of a head

A pan pot changes the level of two channels, and that's all it does. Real ears barely use level to locate a sound. They read three other cues:

- Time. A source on the left reaches the left ear first, by up to three quarters of a millisecond.
- Shadow. The head blocks high frequencies on their way to the far ear, so that ear hears the sound darker.
- The outer ear. Its folds carve notches into the spectrum that move as a source rises. That's how height is heard, and part of what tells front from back.

The usual way to get these cues is to record them, with microphones in the ears of a real or dummy head. Loko computes them instead, from a physical model: a 20 cm sphere for the head and a simplified outer ear.

A model is less literal than a recording, the same way a 3D render is less literal than a photograph. But a camera can only shoot what was in front of it, while a render can show a scene that never existed and still look right, because it follows the rules of light. That trade buys Loko two things a recording can't do:

- Take the ear out and keep the head. A recorded response has the ear's colour baked into every direction. In Loko the outer ear is a stage of its own, so switching **3D** off removes height and the ear's colouring, while left, right and distance stay as they were.
- Bring a source right up to the ear. Recorded responses are almost always taken at one distance, a metre or two away. A sphere has an answer at any distance, so a source at 2 cm sounds like it's *at* the ear, which is a different thing from a loud source further away.

How it works

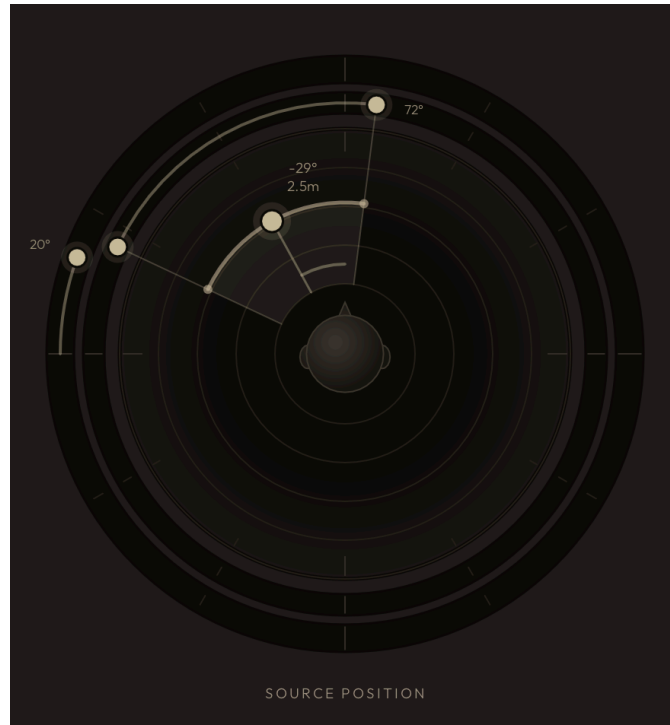
The path from the source to each ear is worked out separately, and each ear gets:

- a delay: the travel time around the head. The difference between the ears is the time cue.
- the head's shadow: the far ear loses top end as the source swings round, and the near ear gains up to 6 dB, as a solid object does on the side facing a sound.
- a near-field response: inside about a metre the near ear gains and the far ear loses, up to about 29 dB apart at the closest distance.
- air absorption, following the ISO standard for sound outdoors: at 200 m a source is 21 dB down at 8 kHz on top of being quieter.
- the outer ear: two resonances and three notches, modelled on the most representative ear in a public database of measured heads rather than on the average ear, since averaging blurs the sharp notches every real ear has.

The level falls 6 dB for every doubling of distance, with 0 dB at 2 m. Last comes a single limiter at 0 dBFS, which applies the same gain to both ears so the left-right balance survives intact. Below full scale it does nothing, and the signal comes out bit for bit.

Loko renders the direct sound only, with no reflections and no tail. The room belongs in a reverb on a send, after Loko.

The placement disc



The pad is the view from above, with the head facing up the screen: the dot's bearing is **Azimuth** and its distance from the head is **Distance**. The outer ring is the vertical plane through that bearing, and its dot is **Elevation** - level with the head at 3 and 9 o'clock, straight up at 12. The inner ring is **Width**, with a dot for each of the source's two copies.

- Hold Shift while dragging on the pad to orbit the head at a fixed distance.
- Drag the elevation dot over the top and the source carries on down the far side, while the dot on the pad swings across the head.
- The rings on the pad sit at 0.02, 0.2, 2, 20 and 200 m, so the close range gets as much room on screen as the far one.

The controls

Azimuth

The bearing, from -180° to $+180^\circ$: 0° is straight ahead, $+90^\circ$ right, -90° left, $\pm 180^\circ$ behind.

Front and back reach the ears at the same times and with the same shadow, so only the outer ear tells them apart, and it does so subtly. To put a source unmistakably behind, place it a little further away too.

Elevation

Height, from -90° (straight down) to $+90^\circ$ (straight up). The outer ear's notches climb from about 8 kHz at ear height to 10 kHz overhead, so a source needs energy up there to be heard rising: a shaker will fly overhead, a bass won't.

Almost every listener hears a generic ear's notches move the right way, but ears vary in size, and only about a third are close enough to the model's to hear exactly the right height. Treat elevation as a dependable relative move and an approximate absolute one.

Distance

From 2 cm to 200 m from the surface of the head. 2 m is the 0 dB reference: closer is louder, further is quieter and duller. Close to the head the level climbs about 45 dB over full scale and the limiter holds it at 0 dBFS, so the last stretch of the dial changes closeness rather than loudness. Lower **Gain** to hear the climb.

Width

How far apart the source's two copies sit, from 0% to 100%. The left input channel goes on one side of the bearing and the right channel on the other, a quarter turn out each at 100%. At 0% they meet in one point. Width is for stereo material: a mono source in two places combs, as one mono track on two speakers does, so leave it at 0% there.

Gain

A level trim from -48 to +48 dB on both ears, ahead of the limiter. It decides how much of the near range the limiter has to hold: at -48 dB even the closest source stays under full scale.

3D

On is the full model. Off removes the outer ear and with it height: Elevation parks at 0° and dims, and comes back when 3D is switched on again during the session. What's left is a strictly two-dimensional head, for a source that only needs placing left and right without a borrowed ear's colouring. The gentle shadow behind the head stays either way, so front and back still differ.

LIMITER

A lamp to watch, with nothing to operate. It glows while the limiter is working, brighter the more it takes off, fully lit at 24 dB. Close sources light it often, which is normal; lowering Gain puts it out.

Latency

A source 200 m away arrives about 583 ms late, and Loko reports zero latency on purpose: that delay is the effect, and compensating for it would undo it. Use Loko as an insert, since a dry copy blended in parallel won't line up.

Tips

- Put the source somewhere a person could stand. Two metres away at 40° left sounds like a place; ten centimetres behind the head sounds like an effect.
- Push a part back with Distance before the fader: it brings the delay and the duller top end along with the drop in level.
- Sources with plenty of top end place best. A pure sine gives the head almost nothing to shadow.

Keyboard and screen readers

Tab and Shift-Tab move between controls, arrow keys nudge the focused one (Shift for finer steps), Return on a dial opens a box to type a value, and Return or Space flips the 3D switch. The disc takes four Tab stops: azimuth, distance, elevation and width. The mouse wheel works on the pad and on both rings.

Loko works with VoiceOver, NVDA, JAWS and Narrator. Every control has a spoken name and value, and the lamp reads its figure in decibels when asked.

Demo and license

There's one download, and it's the full plugin. Until a license is installed the output fades to silence for two seconds every 45 seconds; nothing else is held back.

A purchase comes with a license file by email. Click **DEMO - INSTALL LICENSE** at the foot of the window and pick the file. It's checked on the computer itself, with no connection, account or activation, and it keeps working on a new machine. Licenses are stored in ~/Library/Application Support/StefanoBaldan/Licenses on macOS and in %LOCALAPPDATA%\StefanoBaldan\Licenses on Windows.

Compatibility

Formats: VST3, Audio Unit (AU), CLAP

macOS: 10.15 or later (Apple Silicon and Intel)

Windows: 64-bit

If the DAW doesn't list Loko right after installing, rescan the plug-ins.

Have fun placing things.

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The outer-ear model is derived from the CIPIC HRTF Database, © The Regents of the University of California.

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